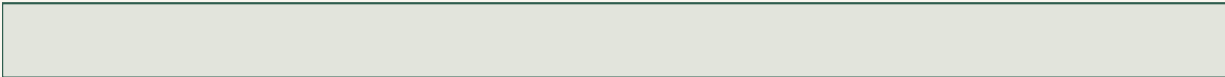


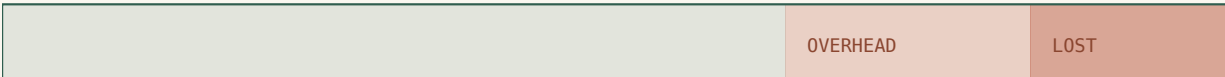
# How to evaluate an AI cloud for large-scale training

*A buyer's guide to the questions that decide whether a cluster finishes your run on time, and what a substantive answer to each one contains.*

ADVERTISED



DELIVERED



The distance between these two bars is the subject of this guide.

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SAMPLE WORK · BUILT ON PUBLIC MATERIAL

# Why the specification sheet cannot answer the question

Two clusters can carry identical accelerators, identical interconnect and identical advertised throughput, and still differ by weeks on the same job. The specification sheet does not capture the difference, because it measures what the hardware can do rather than what it does across a run lasting days.

At scale, training is synchronous. Every device waits on the slowest. When one node fails, the job stops. The cluster rolls back to its last checkpoint, and every unit of computation performed since that moment is discarded.

The loss is not the failure itself. It is the discarded work, plus the time taken to notice, replace and resume.

*Peak throughput tells you how fast a cluster can go. What you are buying is how rarely it stops, and how little you lose when it does.*

## THE THREE NUMBERS THAT MOVE THE OUTCOME

| MEASURE                 | WHAT IT TELLS YOU                                                | WHY IT IS OFTEN ABSENT                                             |
|-------------------------|------------------------------------------------------------------|--------------------------------------------------------------------|
| Model FLOPs utilization | Share of theoretical compute converted into useful training work | Workload dependent, so a single published figure is rarely offered |
| Effective training time | Share of wall-clock time spent making forward progress           | Requires disclosing interruptions                                  |
| Mean time to failure    | How often a run is interrupted at your intended scale            | Scales with node count, so a small-cluster figure misleads         |

Any one of these can be presented favourably in isolation. Requested together, at your intended scale, on a workload resembling your own, they describe the real economics of the run.

## SECTION TWO

# What happens when a node fails

Understanding the failure path is the fastest way to distinguish a mature platform from an immature one, because every stage below is either engineered by the provider or left to you.



### WORTH ASKING

How a failing node is detected, and how quickly. Whether detection is automatic or depends on your own monitoring. Whether a healthy spare is substituted without intervention. Whether nodes are health-checked continuously or only when first provisioned. What restart time has actually been observed at your scale.

### WHAT GOOD LOOKS LIKE

Continuous validation rather than validation at provision time. Pre-emptive removal of degraded nodes before they fail a job. A pool of warm spares held for substitution. Checkpointing guidance tuned to the storage layer. Failure data shared with the customer rather than held.

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*Ask what proportion of node failures the platform detected before the customer noticed. A provider that has instrumented this answers immediately.*

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## SECTION THREE

# The checklist

Take this into the conversation. The second column describes what a substantive answer contains, rather than the answer you are hoping to hear.

| ASK                                                         | A SUBSTANTIVE ANSWER INCLUDES                                                                          |
|-------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| What utilization do you sustain on a workload like ours?    | A named model architecture, a node count and a duration. A figure lacking all three is not comparable. |
| What is your effective training time over a multi-week run? | Measured rather than projected, with the definition of interruption stated.                            |
| What is mean time to failure at our intended scale?         | Figures at your node count, since failure rate scales with fleet size.                                 |
| How is the interconnect provisioned?                        | Topology, whether it is dedicated or shared, and behaviour under contention.                           |
| What does storage sustain during checkpointing?             | Throughput while a full checkpoint is written, not idle benchmark figures.                             |
| What does onboarding look like?                             | Time from contract to first successful run, with a reference customer who will confirm it.             |
| What happens if we scale mid-contract?                      | Lead time for additional capacity and whether it lands in the same fabric.                             |
| What telemetry do we receive?                               | Whether failure and utilization data is exposed to you or retained by the provider.                    |

## THREE ANSWERS THAT SHOULD SLOW YOU DOWN

- i A single utilization figure with no workload attached.** Utilization is workload dependent, so a number without context is marketing rather than measurement.
- ii Benchmarks from a cluster smaller than yours.** Failure behaviour does not scale linearly. A result at 128 nodes says little about 1,024.
- iii No independent validation.** Third-party benchmark submissions and analyst assessments exist in this market. A provider absent from all of them has chosen to be.